

Pellagra

Etiology and Pathogenesis

5-Fluorouracil inhibits the conversion of tryptophan to niacin, and 6-mercaptopurine inhibits NAD phosphorylase, thereby impairing NAD production. Other medications associated with pellagra include phenytoin, chloramphenicol, azathioprine, sulfonamides, and antidepressants.

Clinical Findings

Pellagra is classically characterized by the "four Ds": **dermatitis**, **diarrhea**, **dementia**, and **death**.

- **Dermatitis:** The rash is photosensitive and begins as erythematous, painful, or pruritic patches. Over several days, the skin becomes edematous and may develop vesicles and bullae that rupture, forming crusted erosions, or evolve into brown scales. Chronic changes include sharply demarcated, keratotic, hyperpigmented plaques. - Common sites: dorsum of the hands; when it extends proximally more on the radial than ulnar side, it forms the characteristic "gauntlet" of pellagra. - **Casal's necklace:** A symmetric rash on the upper central chest and neck; may extend over the sternum as a "cravat." - Facial involvement may show a butterfly distribution across the nose, cheeks, chin, and lips. - Palms and soles may develop painful fissures resembling "goose skin."
- Mucous membrane involvement includes cheilitis, angular stomatitis, glossitis, and ulcerations of the buccal mucosa and vulva.
- **Gastrointestinal symptoms:** Often early manifestations—diarrhea, nausea, vomiting, abdominal pain, and anorexia.
- **Neurologic symptoms:** Include insomnia, fatigue, nervousness, apathy, impaired memory, and depression. These may progress to psychosis and dementia in advanced stages.

- **Prognosis:** Untreated pellagra leads to death from multiorgan failure.

Laboratory Testing

Diagnosis is primarily clinical and confirmed by rapid response to vitamin supplementation. Urinary metabolites such as **N-methylnicotinamide** and **pyridone** can be measured to support the diagnosis.

Treatment

- Recommended daily intake: 15–20 mg of niacin or approximately 60 mg of exogenous tryptophan.
- Therapeutic dose: **500 mg/day** of **nicotinamide** or **nicotinic acid** for several weeks.
- **Nicotinamide** is preferred due to fewer side effects (nicotinic acid commonly causes flushing and headache).
- Neuropsychiatric symptoms may improve within 24–48 hours.
- Skin lesions typically resolve over 3–4 weeks.

Vitamin B₆ (Pyridoxine)

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Vitamin **B₆** refers to three interconvertible forms: **pyridoxine**, **pyridoxamine**, and **pyridoxal**. Humans cannot synthesize these compounds and must obtain them from the diet.

- **Dietary sources:** Meats, whole grains, vegetables, and nuts are rich in vitamin B₆.
- **Absorption:** Occurs via passive diffusion in the jejunum.
- **Bioavailability:** Food processing can reduce available vitamin B₆.
- **Active form:** **Pyridoxal-5-phosphate (PLP)** is the most common and physiologically active coenzyme form.